

INTRODUCTION TO BIOLOGY

BIO-101



Chemistry of Life

Lecture 02

Elements

- An element is a substance that **cannot** be broken down into simpler substances by ordinary chemical processes. Every element is made up of its **own type of atom**. This is why the elements are all very different from each other.

- Hydrogen (H): an element

Oxygen (O): an element

Water (H₂O): ???

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 H 1.008																	2 He 4.0026
3 Li 6.94	4 Be 9.0122											5 B 10.81	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
11 Na 22.990	12 Mg 24.305											13 Al 26.982	14 Si 28.085	15 P 30.974	16 S 32.06	17 Cl 35.45	18 Ar 39.948
19 K 39.098	20 Ca 40.078	21 Sc 44.956	22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.38	31 Ga 69.723	32 Ge 72.630	33 As 74.922	34 Se 78.971	35 Br 79.904	36 Kr 83.798
37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.95	43 Tc 98.906	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.91	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm 144.91	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.05		
87 Fr 223.02	88 Ra 226.03	103 Lr 262.11	104 Rf 267.12	105 Db 270.13	106 Sg 269.13	107 Bh 270.13	108 Hs 269.13	109 Mt 278.16	110 Ds 281.17	111 Rg 281.17	112 Cn 285.18	113 Nh 286.18	114 Fl 289.19	115 Mc 289.20	116 Lv 293.20	117 Ts 293.21	118 Og 294.21

57 La 138.91	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm 144.91	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.05
89 Ac 227.03	90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np 237.05	94 Pu 244.06	95 Am 243.06	96 Cm 247.07	97 Bk 247.07	98 Cf 251.08	99 Es 252.08	100 Fm 257.10	101 Md 258.10	102 No 259.10

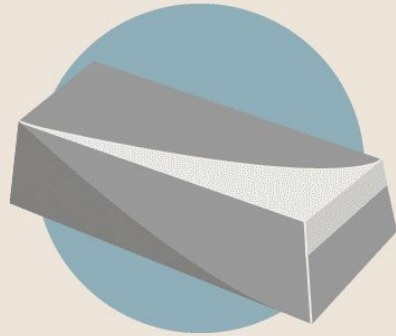
- The periodic table lists all the known elements, grouping together those with similar properties.



Elements (Cont.)

- Most elements are **metals**, some are **non-metals**.

Metals vs. Nonmetals: Physical Properties



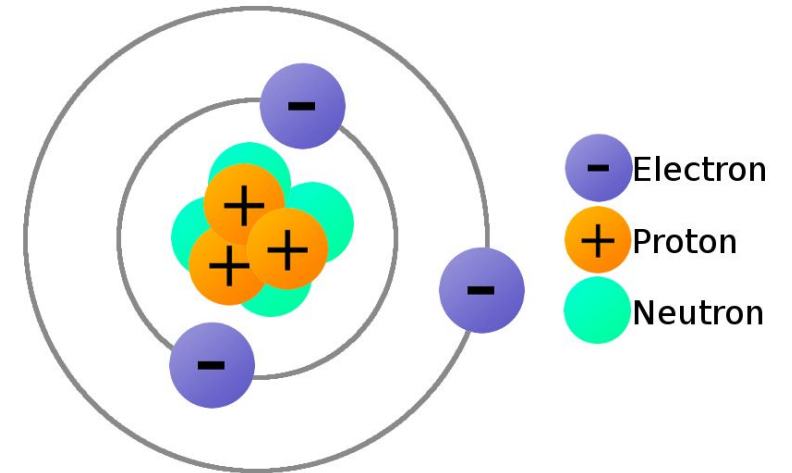
- Lustrous
- Good conductors
- High melting point
- High density
- Malleable
- Ductile (can be drawn into wires)
- Usually solid at room temperature
 - Opaque as a thin sheet
 - Sonorous



- Dull
- Poor conductors
- Nonductile
- Brittle
- May be solids, liquids or gases at room temperature
- Transparent as a thin sheet
 - Not sonorous

What is an atom? What is a molecule?

- Smallest particles that retain properties of an element
- Made up of subatomic particles:
 - Protons (+)
 - Electrons (-)
 - Neutrons (no charge)



- Two or more atoms of any kind, form molecules, when chemically bonded together.

Atomic Number

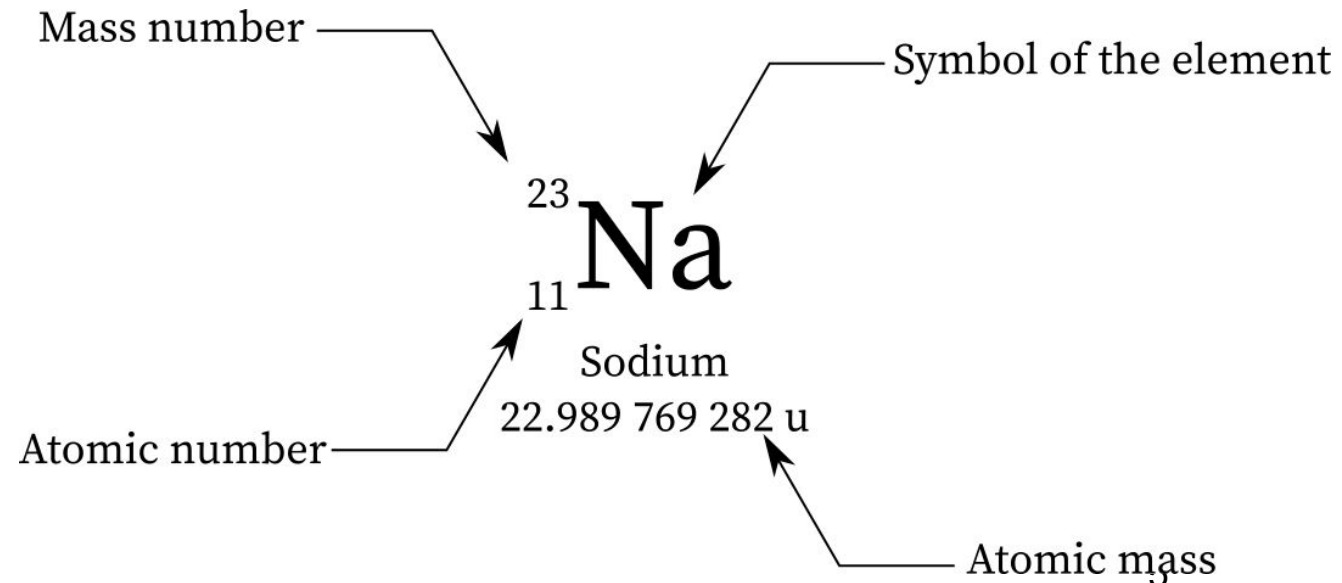
- Number of protons (identity)
- All atoms of an element have the same atomic number
- In the modern periodic table, the elements are listed in order of increasing atomic number.

Atomic Mass

- Atomic mass is the total mass of the atom i.e., the masses of protons, neutrons, and electrons added together.

Mass Number

- Number of protons + number of neutrons
- Isotopes vary in mass number



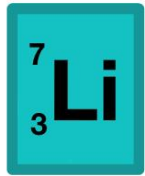
Isotopes, Isobars & Isotones

Isotopes	Isobars	Isotones
Same Atomic Number	Same mass number	Same number of Neutrons
Number of protons & electrons are same. Neutrons only differ	All neutrons, protons, and electrons differ	Number of neutrons are same . Number of electrons & Proton differ
Iso means Same, P stands for Protons	Iso means same, Baros means weight	Iso means same, N means Neutrons

Isotopes



Protons: 3
Neutrons: 3

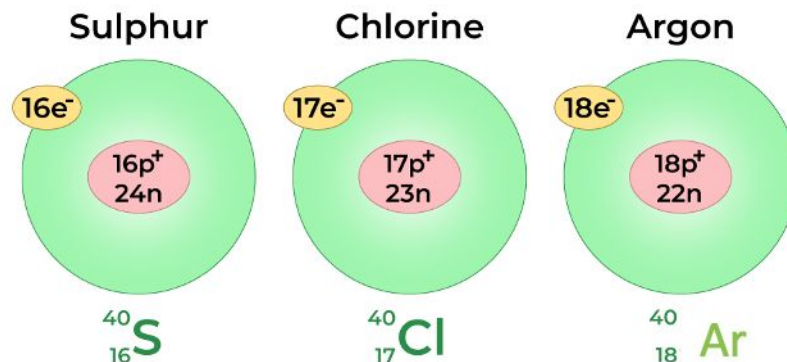


Protons: 3
Neutrons: 4



Protons: 3
Neutrons: 5

Examples of Isobars



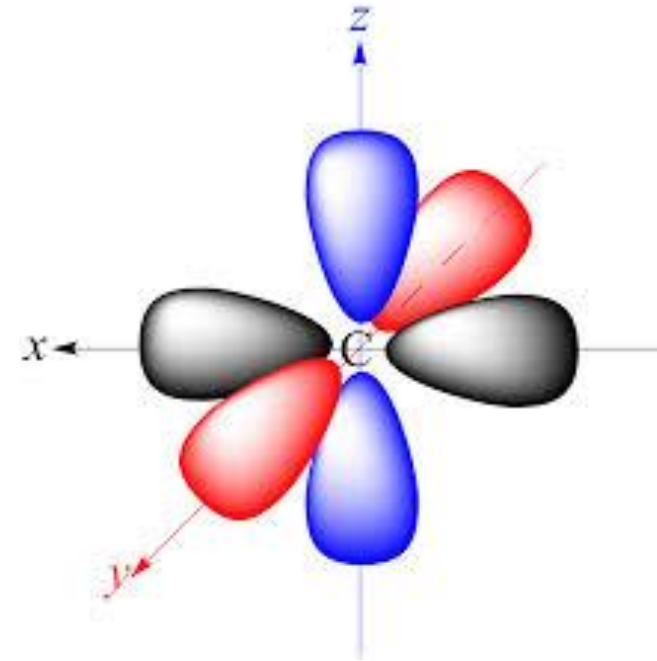
ISOTONES

Example: $^{11}_5\text{B}$ and $^{12}_6\text{C}$

Number of neutrons in boron = $11 - 5 = 6$
 Number of neutrons in carbon = $12 - 6 = 6$

Electrons

- Discovered by British physicist, J. J. Thomson in the year 1897
- Subatomic particle with a negative charge (-)
- Repel one another
- Are attracted to protons in the nucleus
- Move in orbitals, around the nucleus (in the center)



ORBITALS VERSUS ENERGY LEVELS

An orbital is the most probable region where an electron can be found around the nucleus

Orbitals are named as s, p, d, and f

Each orbital can hold a max of 2 electrons. $s=1 \times 2 = 2$,
 $p=3 \times 2 = 6$,
 $d=5 \times 2 = 10$,
 $f=7 \times 2 = 14$

Energy levels are somewhat fixed distances from the nucleus, where electrons may be found.

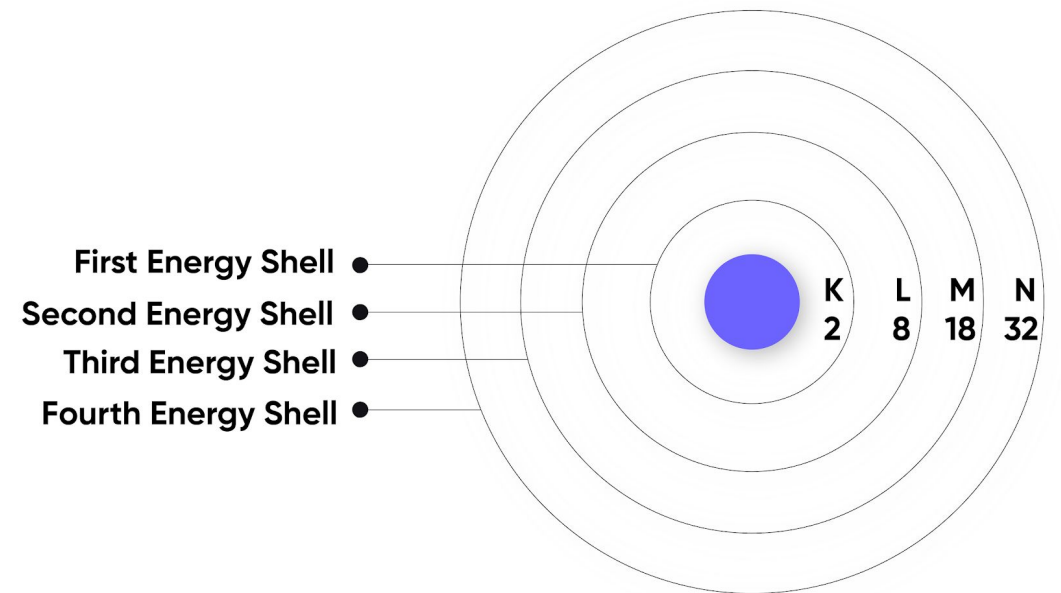
Energy levels are named as K, L, M, N

The number of electrons, each energy level can hold can be determined by $2n^2$, where “n” is the number of the energy level, i.e. 1, 2, 3 and so on.

Visit www.pediaa.com

Energy Levels/ Shells

- Generally, each energy level or shell contains a maximum of $2n^2$ electrons where, n is the number of the shell. By just substituting the shell number in “ n ” we can easily calculate number of electrons in each shell.
- First shell occupies a maximum of two electrons i.e., $2 \times 1^2 = 2$,
second shell occupies a maximum of eight electrons i.e., $2 \times 2^2 = 8$,
third shell occupies a maximum of 18 electrons i.e., $2 \times 3^2 = 18$ and so on...



Chemical Bonds

- A chemical bond is a lasting attraction between atoms or ions that enables the formation of molecules.
- When a chemical bond is formed, the structure and characteristics of atoms don't change.

Types of Chemical Bonds

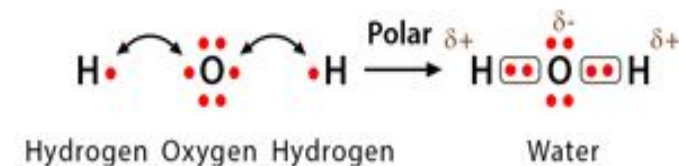
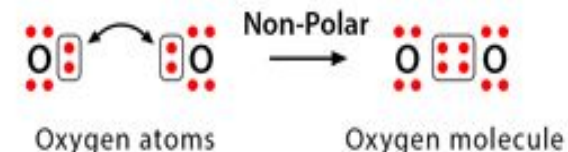
1. Ionic Bond

Metal atom loses electron(s) to nonmetal atom



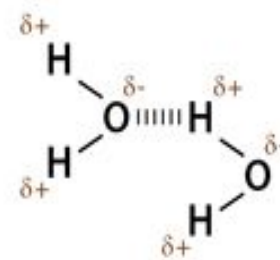
2. Covalent Bond

Two nonmetal atoms share electrons



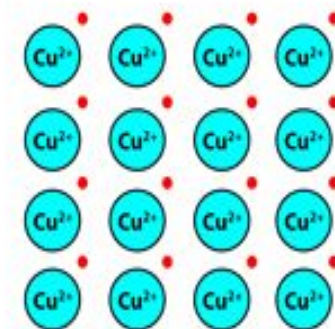
3. Hydrogen Bond

Hydrogen attracts an electronegative atom electrostatically



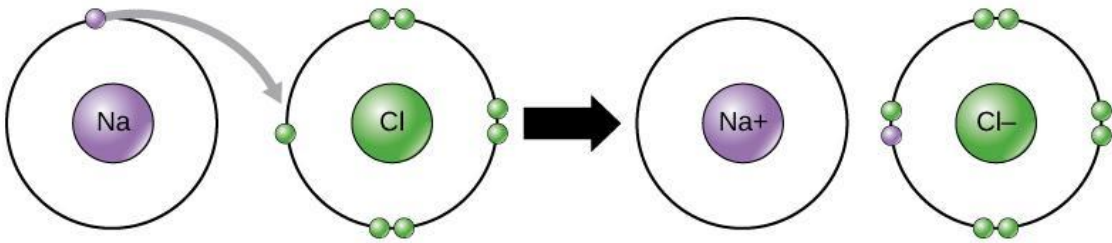
4. Metallic Bond

Positive metal ions attract conducting electrons



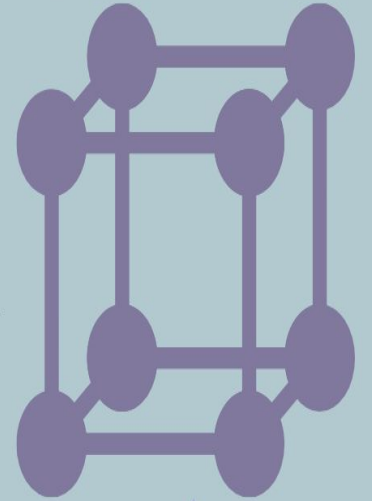
IONIC BOND

- Ionic bond forms when valence electron(s) are completely transferred between atoms.
- In this bond, metal atom loses electron(s), becomes positively charged ion whereas a nonmetal atom gains the electron(s), becomes negatively charged ion.
- Charge difference attracts the two ions to each other.
- NaCl, KCl.



Properties of Ionic Compounds

- Form crystals
- Hard and brittle
- High melting and boiling points
- Don't conduct electricity as solids, but do conduct when molten or dissolved in water
- Often soluble in water and insoluble in nonpolar solvents
- High enthalpies of fusion and vaporization
- Solids are good insulators



COVALENT BOND

- Occurs when pair(s) of electrons are shared between atoms.
- The shared electrons located between the two nuclei are: shared pair
- EXAMPLES: H_2O , CH_4 etc.
- There are two types of covalent bonds-
 - a. Polar Covalent bond
 - b. Non-polar Covalent bond

Properties of Covalent Compounds

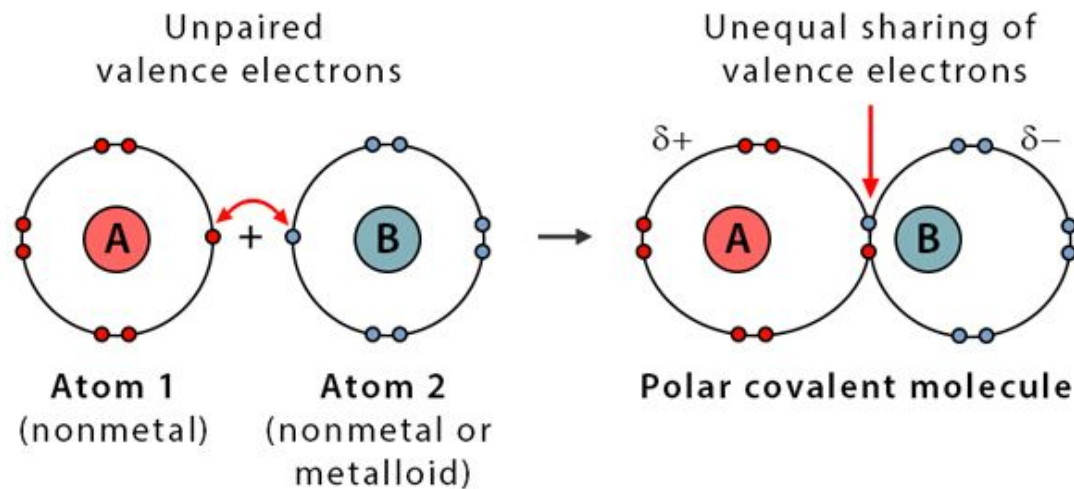
- Covalent bonds are weaker than ionic or metallic bonds
- Small covalent compounds tend to be gases, with relatively weak intermolecular forces
- Larger molecules have greater attractive forces; can be liquids or solids
- Polar molecules have considerably higher melting and boiling points than non-polar molecules

Depending on the polarity that is creation of positive and negative poles, covalent bonds can be of two types.

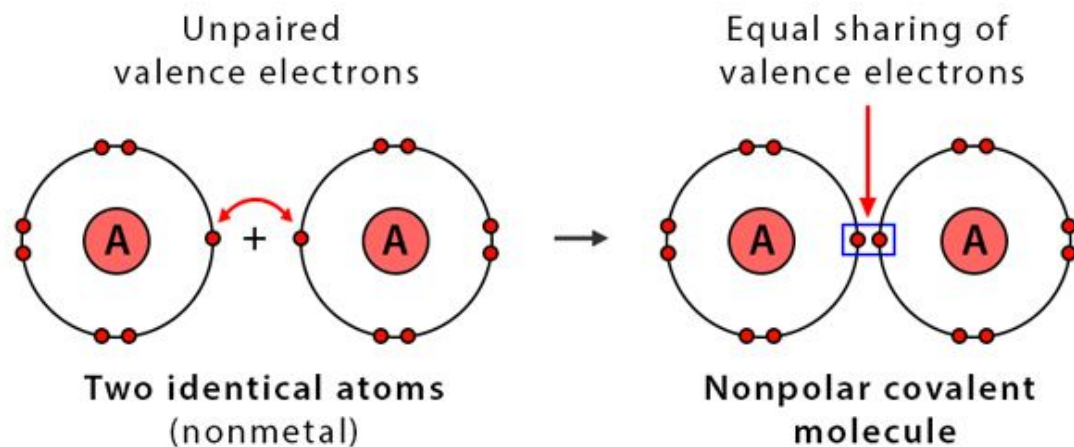
- **A polar covalent bond** forms when number of protons in nuclei of participating atoms is NOT equal (atoms with different electronegativities).
 - Electrons spend more time near nucleus with most protons
 - For e.g. HCl
- **A nonpolar covalent bond** forms when electrons are shared equally between two atoms.
 - Nuclei of atoms have same number of protons
 - For e.g. H₂

Polar vs Nonpolar Covalent Bond

Polar Covalent Bond



Nonpolar Covalent Bond



POLAR COVALENT BONDS

Polar covalent bonds are chemical bonds where electrons are shared unequally between two atoms due to the difference in electronegativity

Compounds with polar covalent bonds are soluble in water

Compounds with polar covalent bonds exhibit polarity

Polar covalent bonds possess a dipole moment

NONPOLAR COVALENT BONDS

Nonpolar covalent bonds are chemical bonds in which electrons are shared equally between two atoms due to similar electronegativities

Compounds with nonpolar covalent bonds are not soluble in water

Compounds with nonpolar covalent bonds do not have polarity

Nonpolar covalent bonds have a dipole moment equal to zero

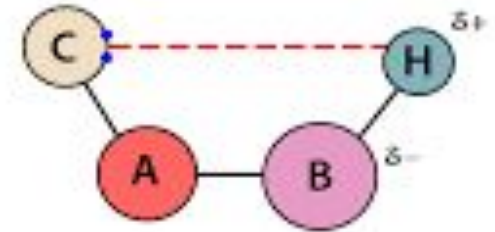
Hydrogen Bond

Hydrogen bonds are created between hydrogen (could be of a polar molecule) and highly electronegative molecules.

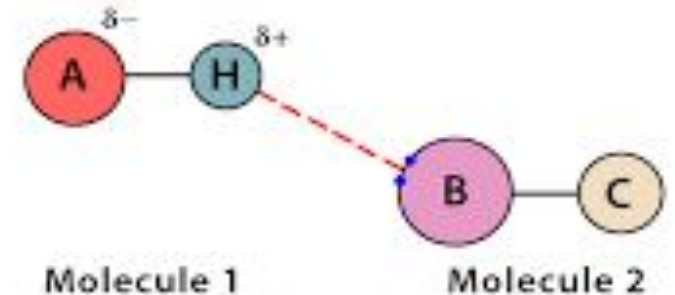
- Molecules held together by polar covalent bonds has no NET charge
- However, atoms of the molecule carry different charges
- Atom in one polar covalent molecule can be attracted to oppositely charged atom in another such molecule

Types of Hydrogen Bond

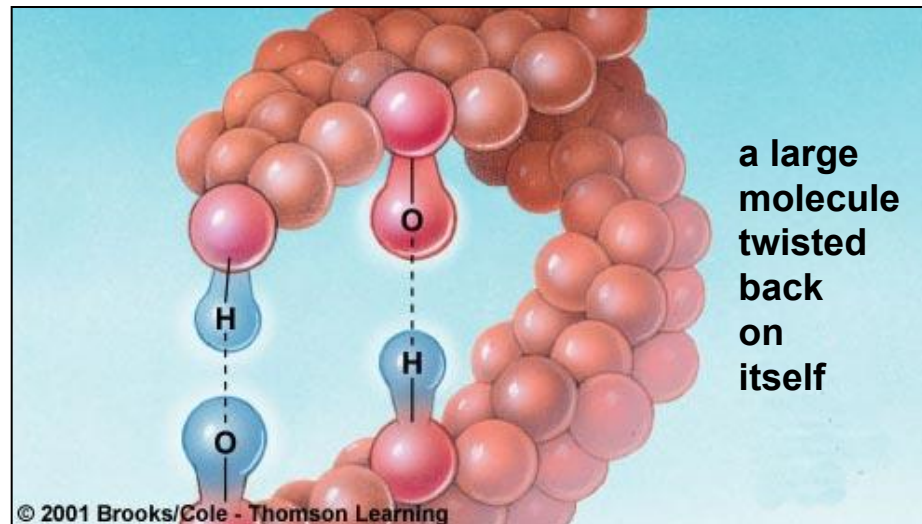
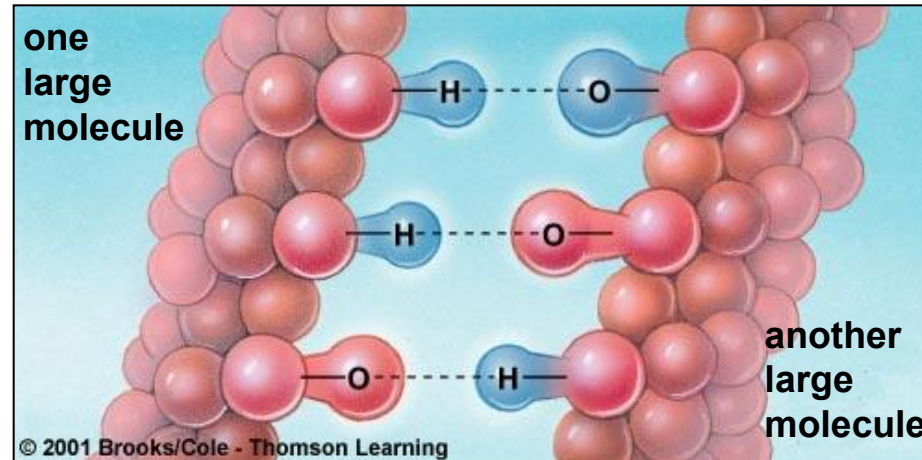
1. Intramolecular



2. Intermolecular

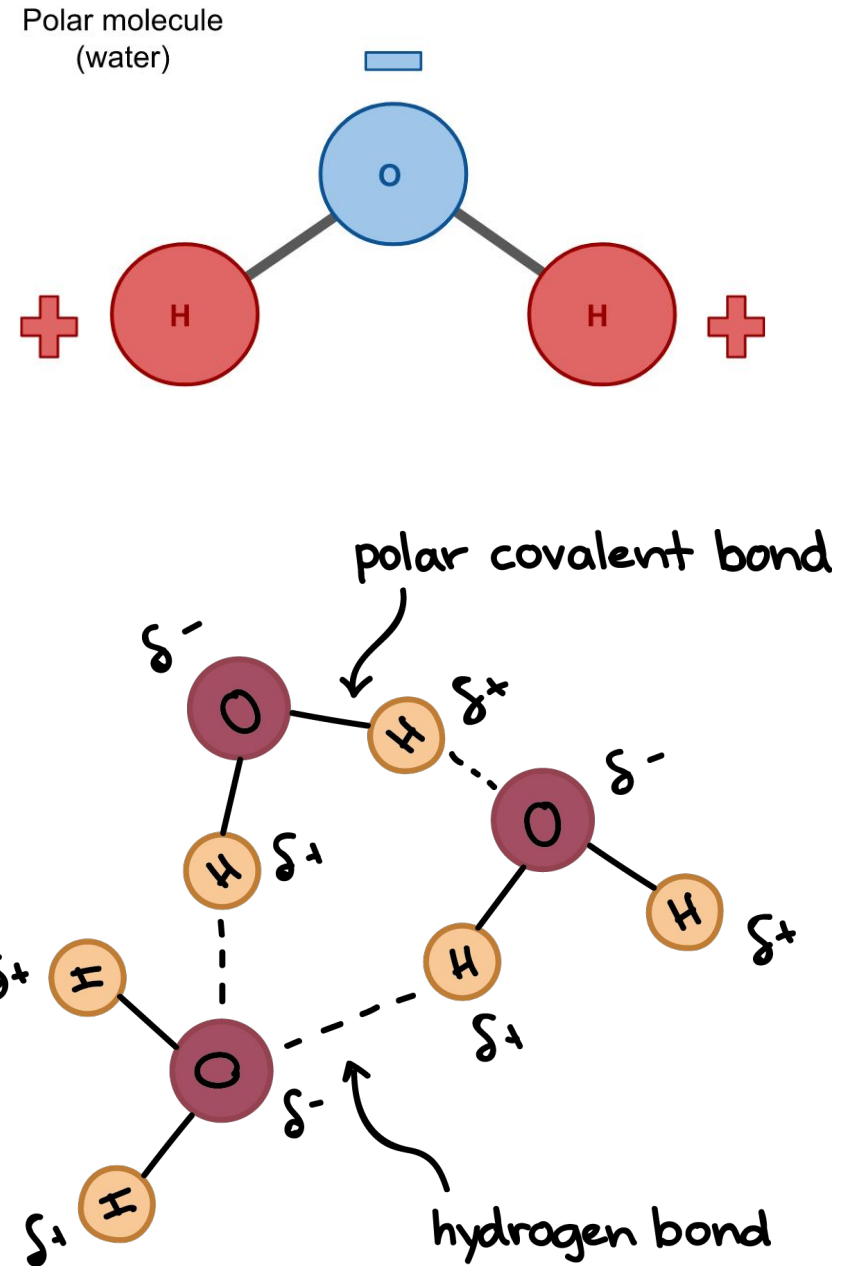


Examples of Hydrogen Bonds



Hydrogen bond in water

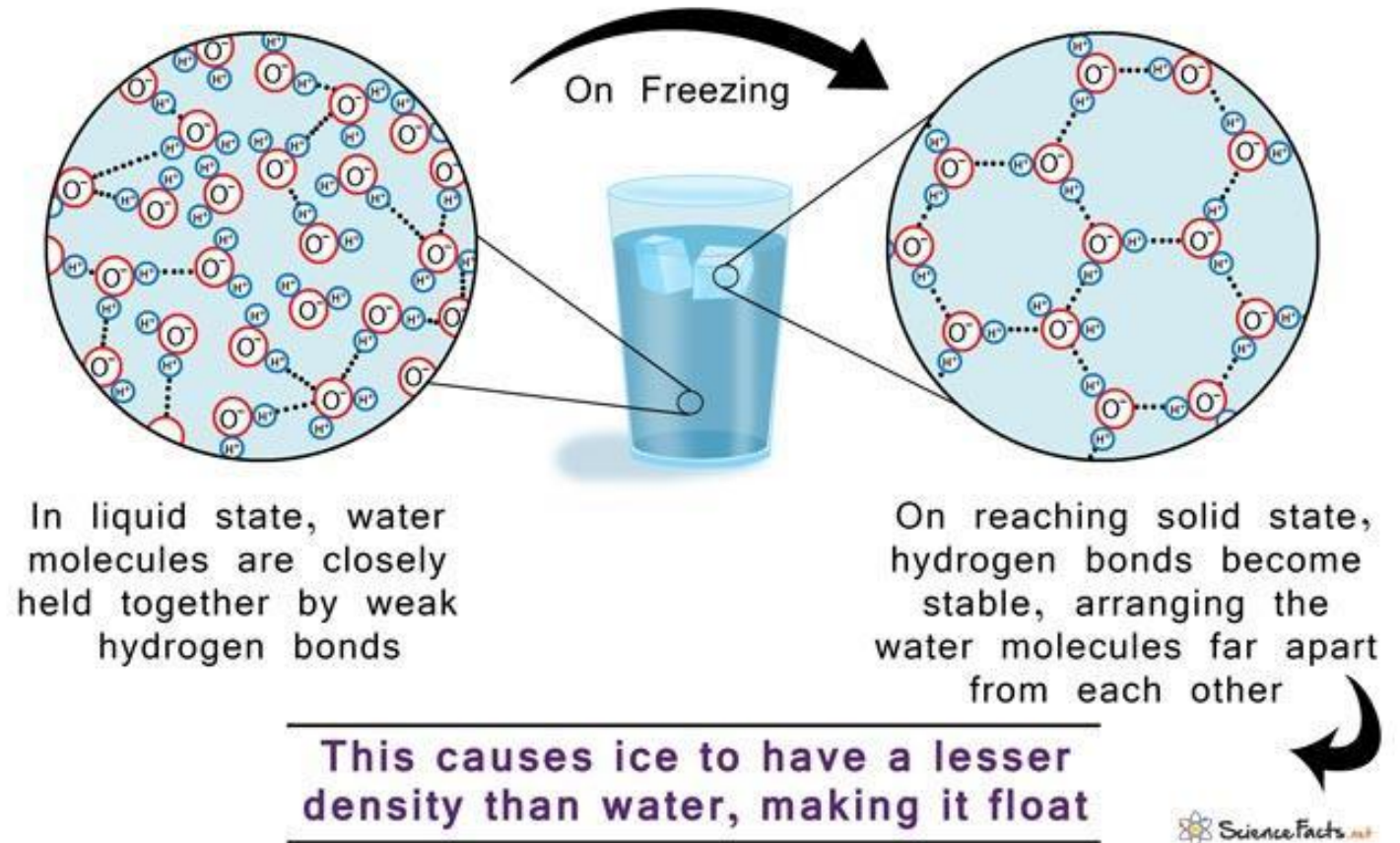
- Water is a polar molecule
- It has NO NET CHARGE
- Oxygen has a strong attraction for electrons than hydrogen.
- They pull the shared electron towards them → form a partial negative charge.
- Hydrogen end → slight positive charge
- Which results into hydrogen bond formation between neighboring hydrogen and oxygen atoms of adjacent water molecules.



Why Ice Floats

- In ice, hydrogen bonds lock molecules in a lattice - open type structure with wide holes
- H_2O molecules in lattice are spaced farther apart than those in liquid water
- Ice is less dense than water
→ floats on water

WHY DOES ICE FLOAT ON WATER



Importance of Hydrogen Bond in Biology

- Source of unique properties of water
- Structure and function of proteins
- Structure and function of DNA
- Structure and function of polysaccharides
- Binding of a substrates to enzymes
- Binding of hormones to receptors
- Matching of mRNA and tRNA



Hydrophilic & Hydrophobic Substances

- Hydrophilic substances
 - Polar
 - form hydrogen bond with water
 - Example: sugar



- Hydrophobic substances
 - Nonpolar
 - Repelled by water
 - Example: oil

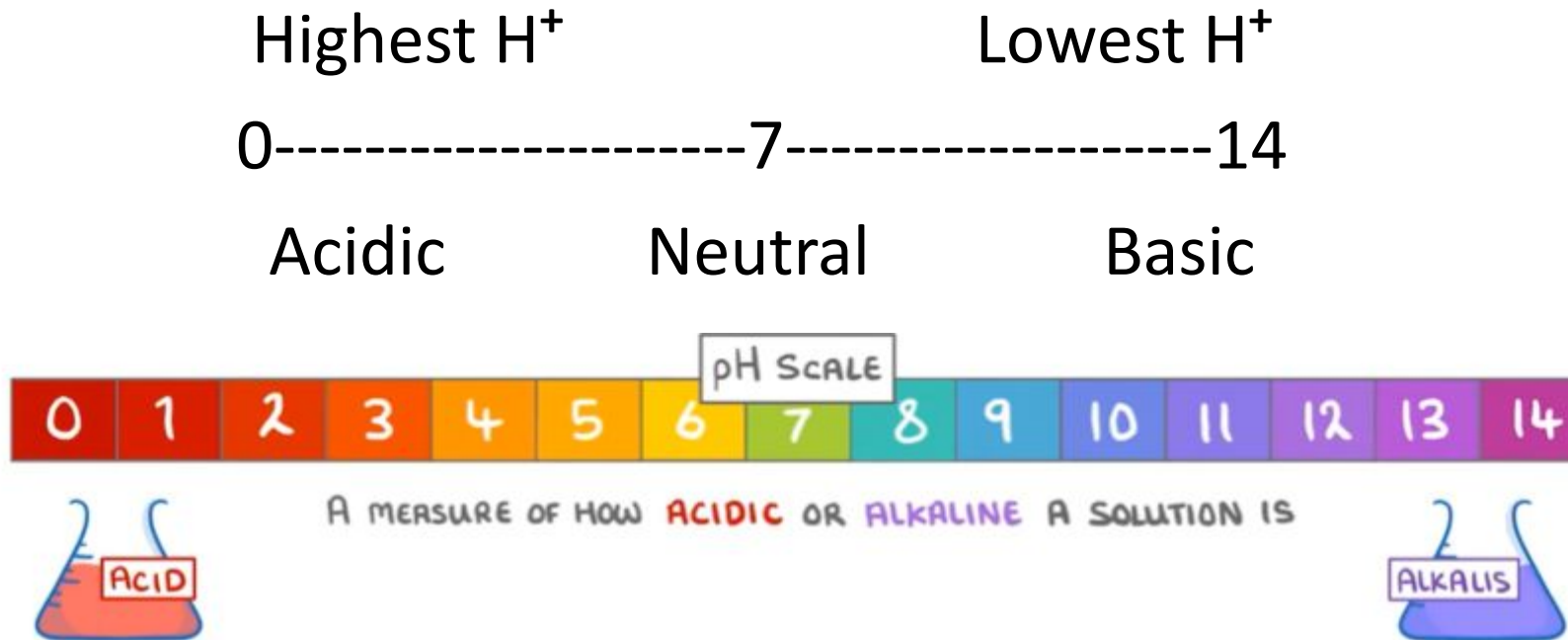


The pH Scale

- Measures H^+ concentration of fluid -> how acidic/ basic a solution is
- Logarithmic scale which ranges from 0 to 14, 7 being neutral

pH = -log [H⁺]

- Change of 1 on scale means 10X change in H^+ concentration



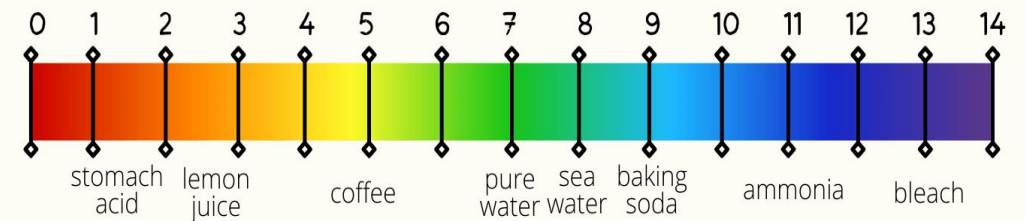
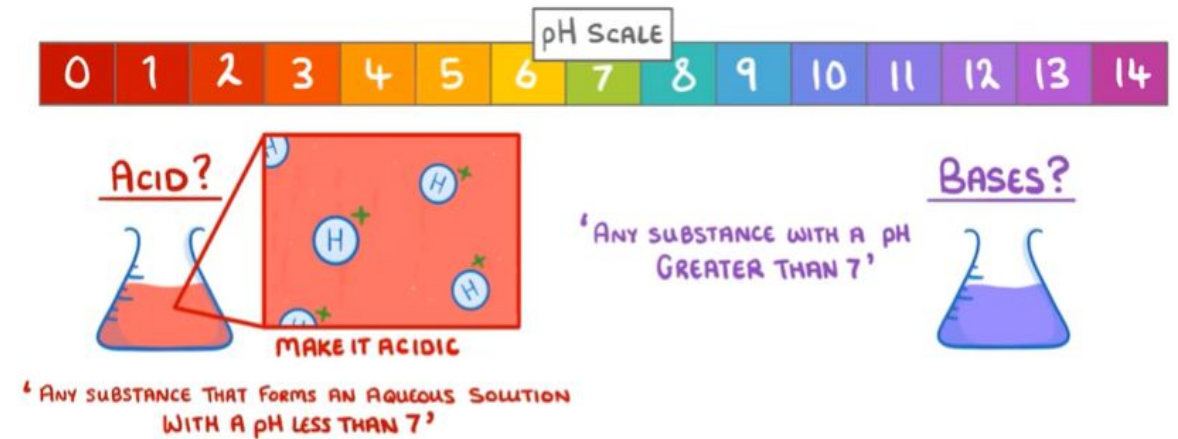
Acids & Bases

- Acids

- Donate H^+ when dissolved in water
- Acidic solutions have $pH < 7$
- Example: HCl

- Bases

- Accept H^+ when dissolved in water
- Basic solutions have $pH > 7$
- Example: NH_3



acids

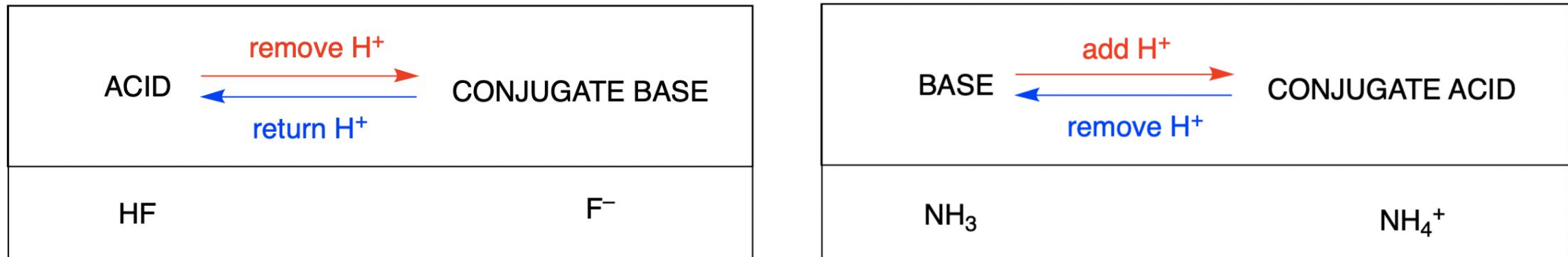
- pH less than 7
- Turn litmus paper red
- Taste sour
- Sting or just feel wet
- Include fruit juices, soda, and coffee

bases

- pH greater than 7
- Turn litmus paper blue
- Taste bitter or soapy
- Feel slippery
- Include baking soda, ammonia, and soap

Buffer System or Buffer Solution

- Its pH changes very little when a small amount of strong acid or base is added to it.
- Purpose: To keep pH at a nearly constant value
- Aqueous solution of a weak acid + its conjugate base, or a weak base + its conjugate acid.



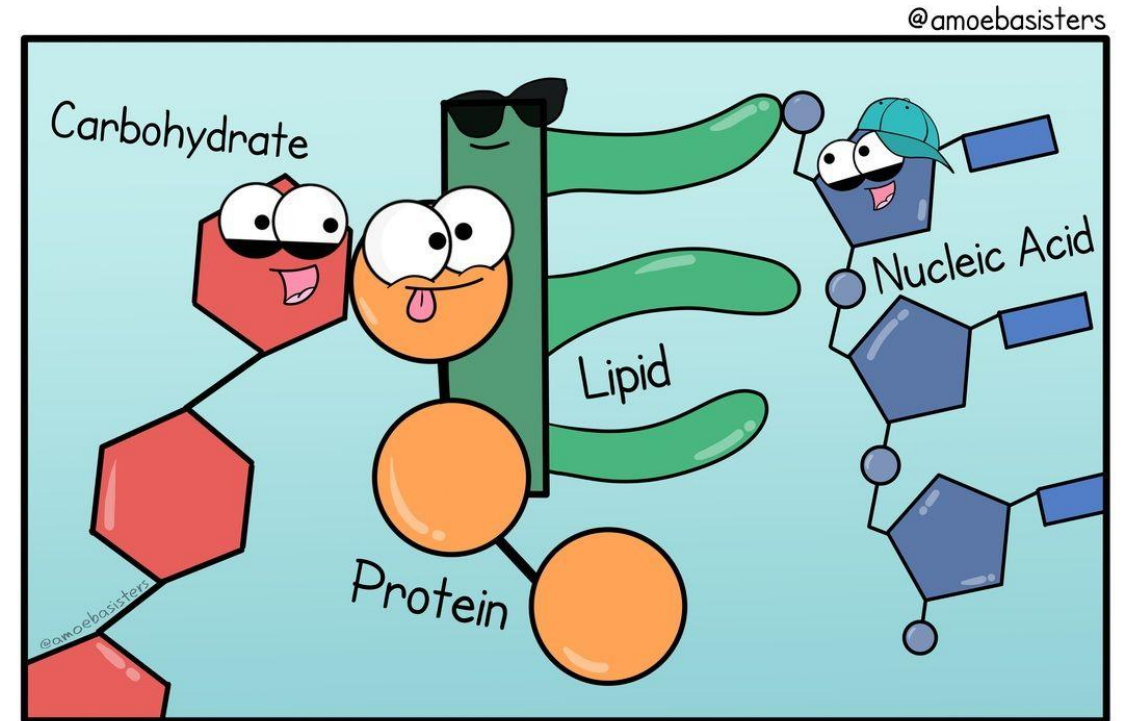
- One example of a buffer solution found in nature is blood: carbonic acid H₂CO₃ & bicarbonate ion HCO₃⁻ system.
The body's acid–base balance is normally tightly regulated, keeping the blood pH between 7.38 and 7.42.

Types of Buffer Solutions

- Buffers are broadly divided into two types: acidic and alkaline buffer solutions.
- **Acidic buffers** are solutions that have a pH below 7 and contain a weak acid and one of its salts. For example, a mixture of acetic acid and sodium acetate acts as a buffer solution with a pH of about 4.75.
- **Alkaline buffers**, on the other hand, have a pH above 7 and contain a weak base and one of its salts. For example, a mixture of ammonium hydroxide and ammonium chloride acts as a buffer solution with a pH of about 9.25.

What are Biomolecules?

- Biomolecules are molecules present in organisms that are essential to one or more biological processes.
- Most biomolecules are **organic compounds** containing four major elements e.g. carbon, hydrogen, oxygen, and nitrogen.
- Being “organic” means, in general, they contain carbon atoms covalently bound to other atoms, especially Carbon-Carbon (C-C) and Carbon-Hydrogen (C-H).
- Many of the biomolecules are polymers: compounds made up of several repeating units (monomers)



BIOMOLECULE BROS!

Carbohydrates

- . A carbohydrate is a biomolecule consisting of carbon (C), hydrogen (H) and oxygen (O) atoms.
- . Saccharide is the monomer of carbohydrates. Based on the number of monomers, carbohydrates may be monosaccharides, disaccharides, oligosaccharides, or polysaccharides.
- . Carbohydrates perform numerous roles in living organisms.
- . They can store energy (e.g. starch and glycogen), act as structural components (e.g. cellulose in plants) and even form the backbone of the genetic molecule, nucleic acids (RNA & DNA)



Classification of carbohydrates



Glycosidic Bond

- . A glycosidic bond is a covalent bond formed between a carbohydrate molecule and another molecule.
- . In this reaction, the hydroxyl group (OH) of the carbohydrate combines with the hydrogen of another organic molecule, releasing a molecule of water and forming a covalent bond.

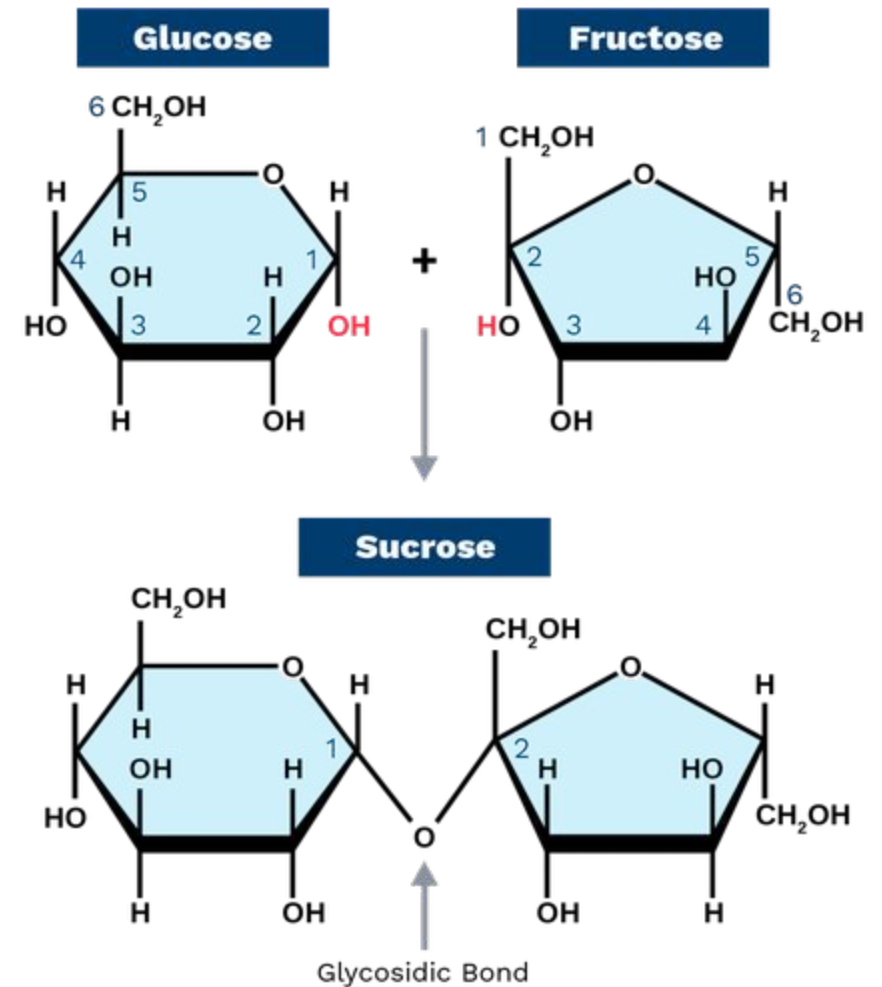


Fig: Sucrose is formed when a monomer of glucose and a monomer of fructose are joined in a **dehydration reaction**

Lipids

- Lipids are biomolecules that contain hydrocarbons and make up the basic building blocks of biological membranes. They have various function in living cells.
- Lipids are not soluble in water as they are non-polar, but are soluble in non-polar solvents such as chloroform.



· Examples of lipids include fats, oils, waxes, hormones etc.

· The functions of lipids include storing energy, signaling, and acting as structural components of cell membranes.

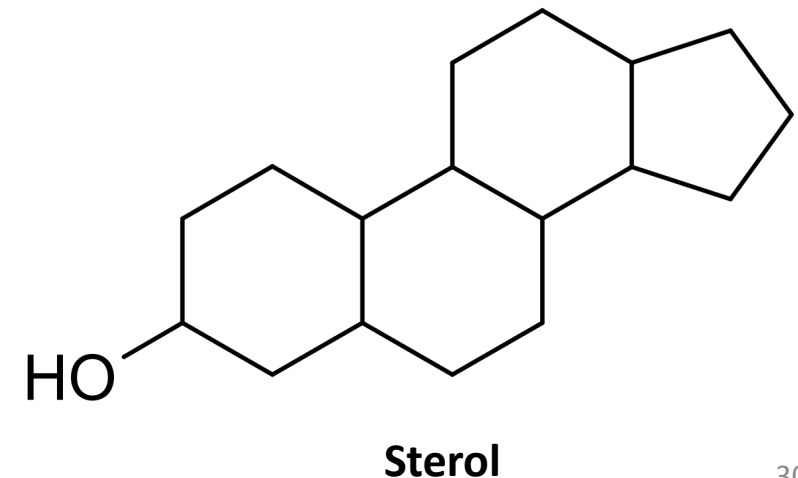
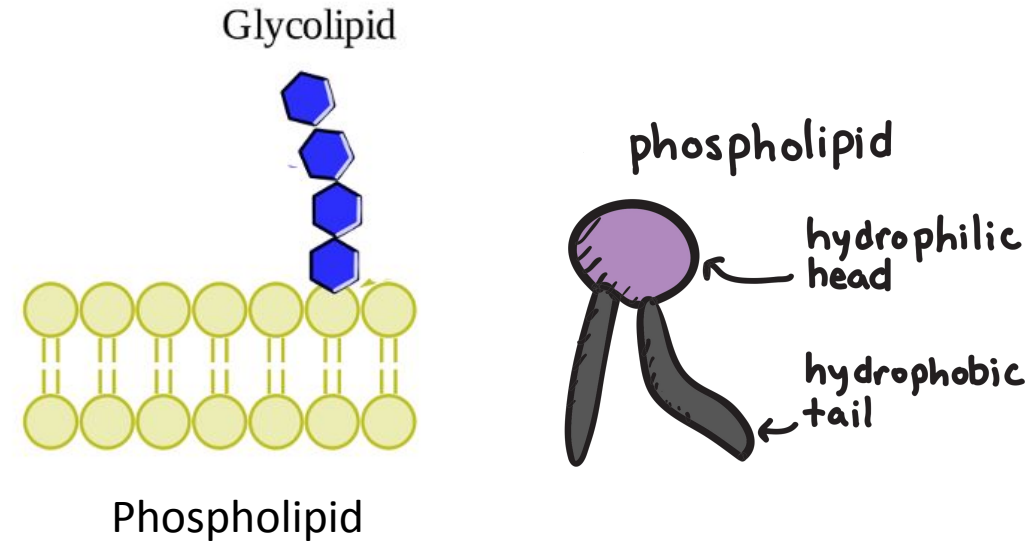
Major Types of Lipids

Triglycerides are derived from glycerol and three fatty acids. Main constituents of body fat, vegetable fat, also oil.

Glycolipids are a class of lipids containing carbohydrate residues. Their role is to maintain the stability of the cell membrane and to facilitate cellular recognition.

A **phospholipid** is a type of lipid molecule that is the main component of the cell membrane -> Bilayer phospholipids

Sterols are unique among lipids in that they have a multiple-ring structure. They contain a planar (flat) steroid ring, for example- cholesterol.



Ester Bond

- Triglyceride is basic unit of lipids
- Glycerol is a small organic molecule with three hydroxyl (OH) groups, while a fatty acid consists of a long hydrocarbon chain attached to a carboxyl group.
- A triglyceride is formed when 1 glycerol molecule links with 3 fatty acid molecules by means of ester bond (covalent bond).
- This process releases 3 water molecules.

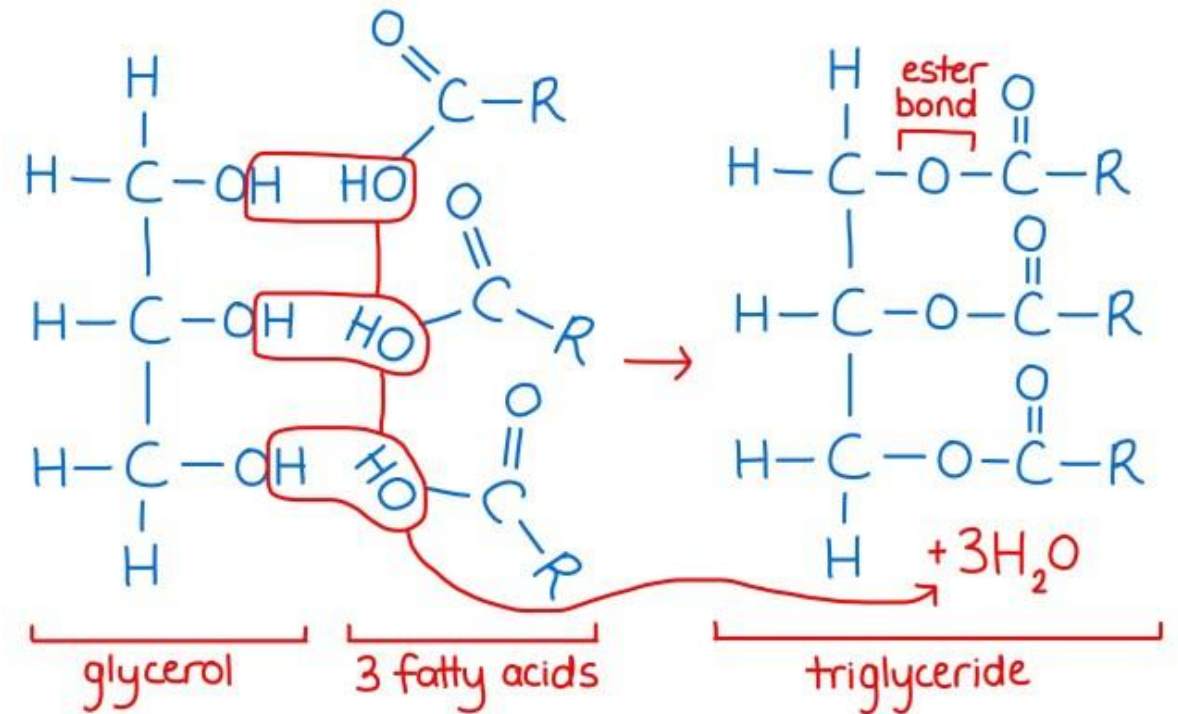


Fig: The glycerol molecule (having three OH- groups) links with three fatty acids via an ester bond

Saturated and Unsaturated Fats

Saturated Fat	Unsaturated Fat
meats, butter, dairy products	vegetable oils
solid at room temperature	liquid at room temperature
increase levels of “bad” cholesterol (low-density lipoprotein)	increase levels of “good” cholesterol (high-density lipoprotein)
low-density lipoprotein clogs arteries	high-density lipoprotein, or HDL, “grabs” LDL and escorts it to the liver where LDL is broken down and eventually removed from the body



Saturated	Unsaturated
$\begin{array}{c} \text{H} \text{ H} \\ \quad \\ -\text{C}-\text{C}- \\ \quad \\ \text{H} \text{ H} \end{array}$	$\begin{array}{c} \text{H} \text{ H} \\ \quad \\ -\text{C}=\text{C}- \end{array}$
Carbon-Carbon Single Bond	Carbon-Carbon Double Bond

Proteins

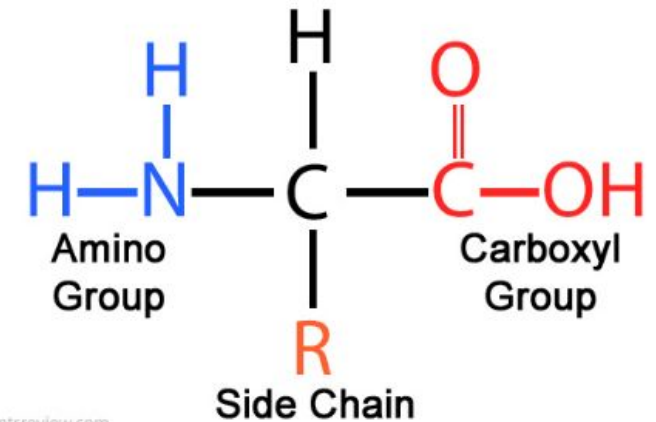
Proteins are large biomolecules or macromolecules that comprise one or more long chains of **amino acid residues**.

Proteins within a cell have many functions: providing structural support, building cellular structures

They help in metabolism by acting as enzymes, carriers and hormones.



Amino Acid Structure



Peptide Bond

Within a protein, multiple amino acids are linked together by peptide bonds, thereby forming a long chain.

A peptide bond is a chemical bond formed between two amino acids when the carboxyl group of one molecule reacts with the amino group of the other molecule, releasing a molecule of water (H₂O).

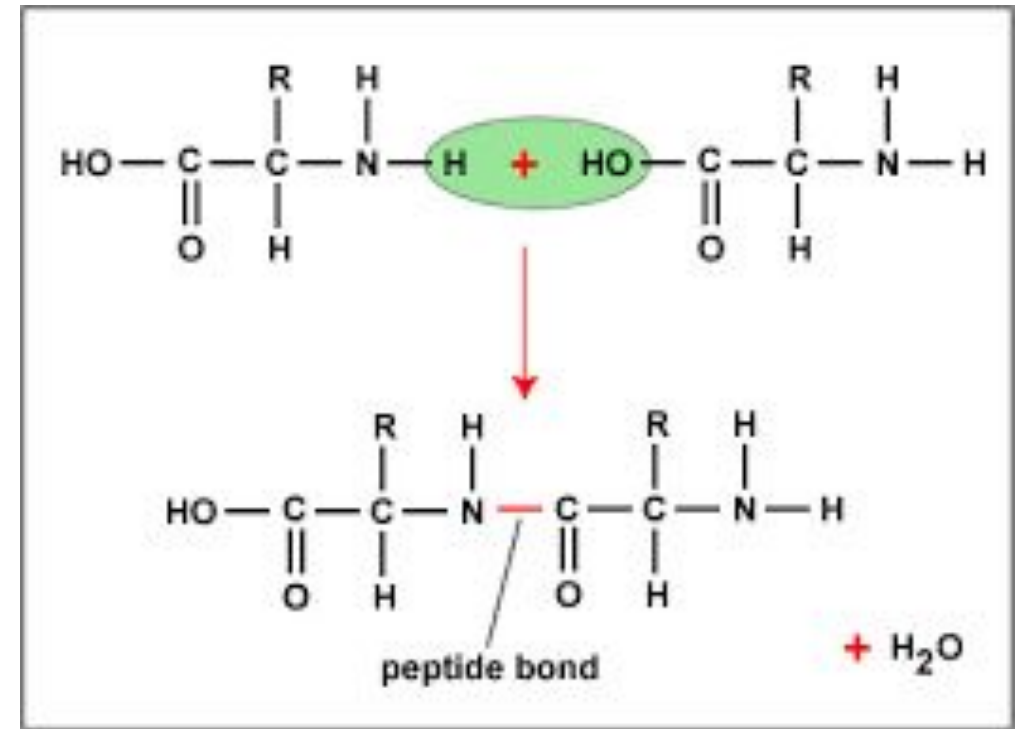


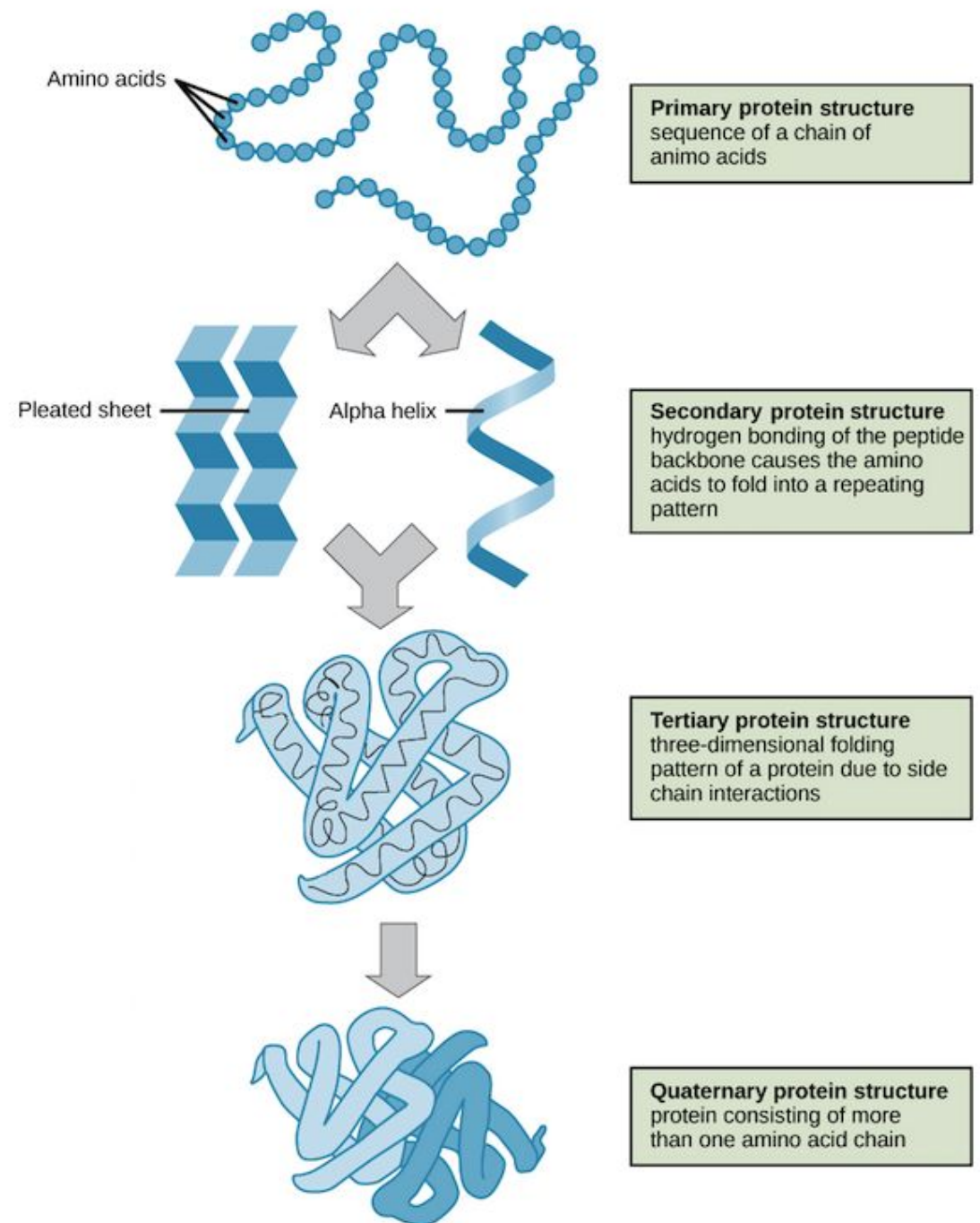
Fig- Formation of a peptide bond

PRIMARY VS SECONDARY VS TERTIARY

STRUCTURE OF PROTEIN

PRIMARY STRUCTURE	SECONDARY STRUCTURE	TERTIARY STRUCTURE
Linear sequence of amino acids	Folding of the peptide chain into an α -helix or β -sheet	Three-dimensional structure of a protein
Linear	Either an α -helix or β -sheet	Globular
Composed of peptide bonds formed between amino acids	Encompasses hydrogen bonds	Encompasses disulfide bridges, salt bridges, and hydrogen bonds
Formed during translation	Forms collagen, elastin, actin, myosin, and keratin-like fibers	Includes enzymes, hormones, albumin, globulin, and hemoglobin
Involved in post-translational modifications	Involved in forming structures such as cartilages, ligaments, skin, etc.	Involved in the metabolic functions of the body

Visit www.pediaa.com



Protein Shapes

Fibrous proteins

typically elongated and insoluble in water.
provide structural support

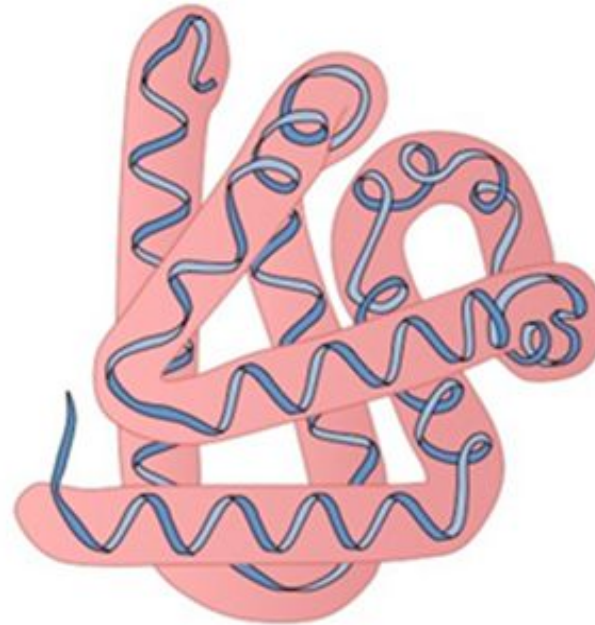
has repetitive amino acid sequence.

less sensitive to heat/ pH

Examples - collagen, found in the bones, muscles, skin



Fibrous Protein



Globular Protein

Globular proteins

generally compact, spherical in shape and soluble in water.
provide functional support (catalyst, transport).

has irregular amino acid sequence.

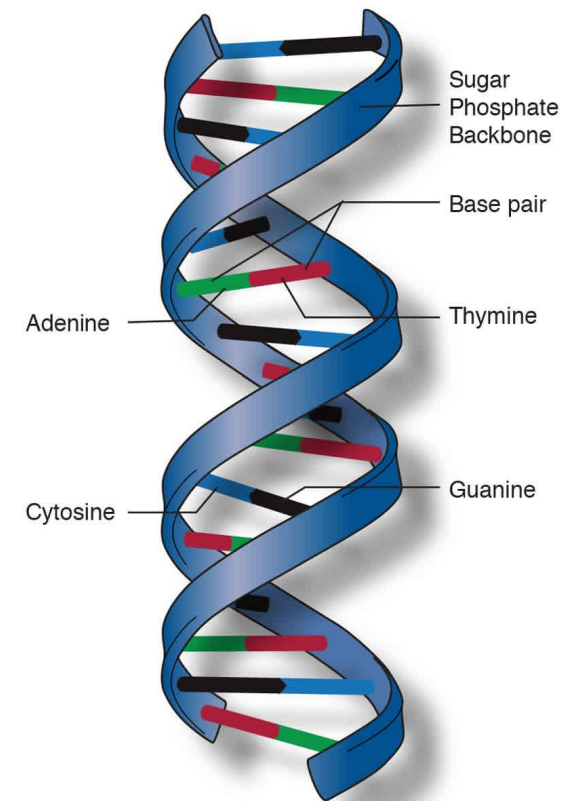
more sensitive to heat/ pH

Examples – hemoglobin, insulin

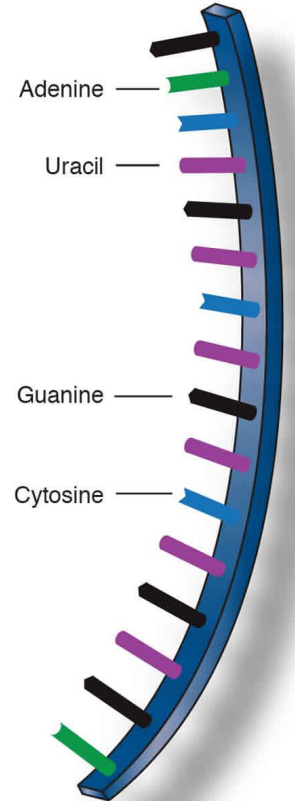
Fig: Fibrous and globular proteins

Nucleic Acids

- Nucleic acids are essential for all forms of life, and it is found in all cells.
- The functions of nucleic acids have to do with the storage and expression of genetic information.
- The building blocks of nucleic acids are nucleotides, bonded by phosphodiester bonds.



Deoxyribonucleic acid
(DNA)



Ribonucleic acid
(RNA)

Deoxyribonucleotides

Nucleotides that make up DNA are called **deoxyribonucleotides**.

The three components of a deoxyribonucleotide are a five-carbon sugar called **deoxyribose**, a **phosphate group** and a **nitrogenous base** (Adenine/ **Thymine**/ Guanine/ Cytosine).

DNA stores the information needed to build and control the cell.

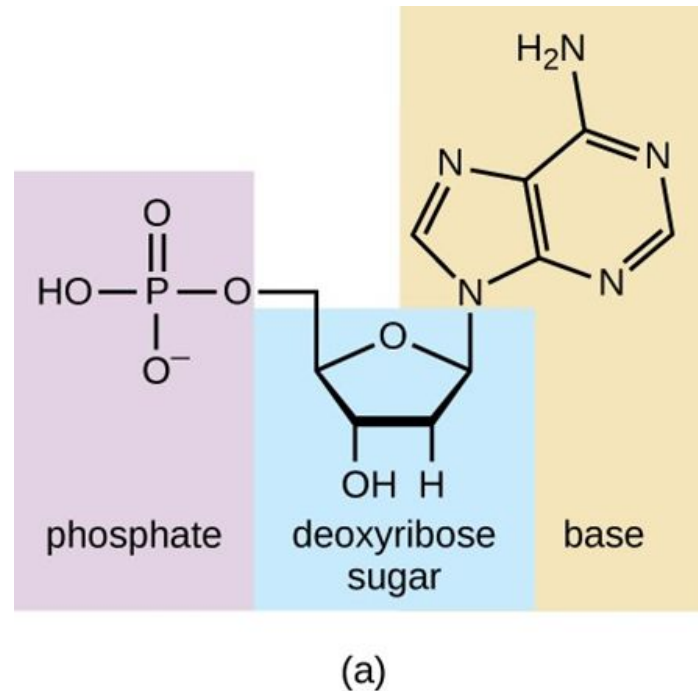
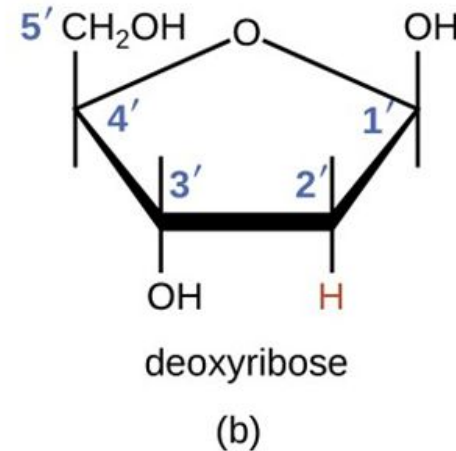


Fig: (a) deoxyribonucleotide



(b) deoxyribose sugar

Ribonucleotides

- Nucleotides that make up RNA are called ribonucleotides. The three components of a ribonucleotide are a (5 Carbon) **pentose sugar called ribose**, a phosphate group and a nitrogenous base (Adenine/ **Uracil**/ Guanine/ Cytosine).
- The RNA-specific pyrimidine **uracil** forms a complementary base pair with adenine (a replacement of **thymine** in DNA).
- The primary role of RNA is to convert the information stored in DNA into proteins.

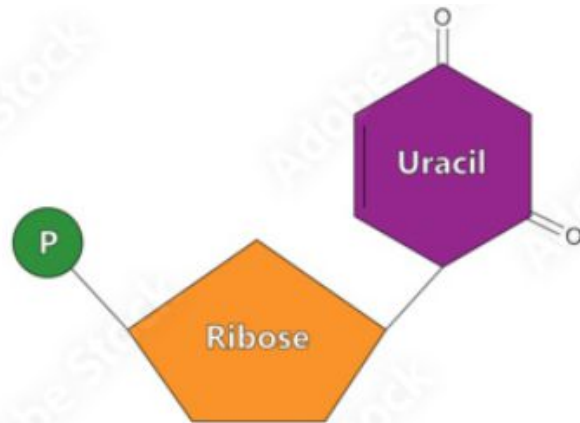
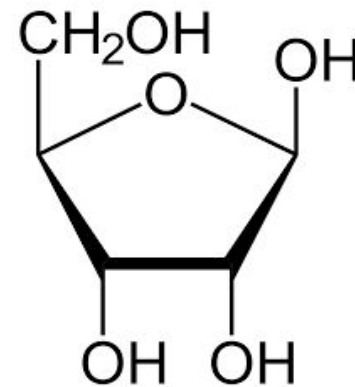


Fig: (a) ribonucleotide,



(b) ribose sugar

Nitrogenous Bases

The nitrogenous bases adenine (A) and guanine (G) are the **purines**; they have a double-ring structure with a six-carbon ring fused to a five-carbon ring.

The **pyrimidines**, cytosine (C) and thymine (T), are smaller nitrogenous bases that have only a six-carbon ring structure. **Thymine is unique to DNA, Uracil is unique to RNA!**

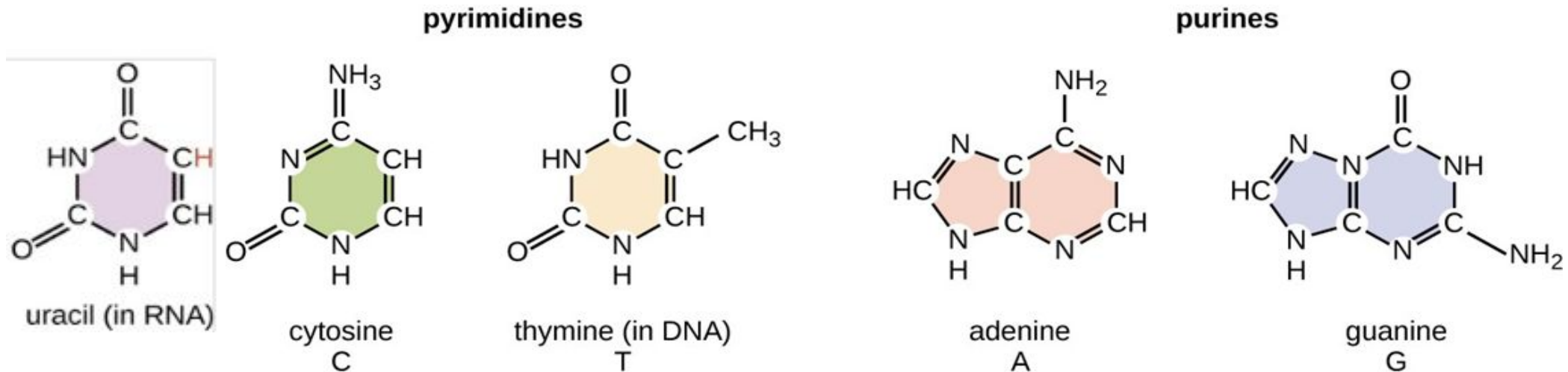


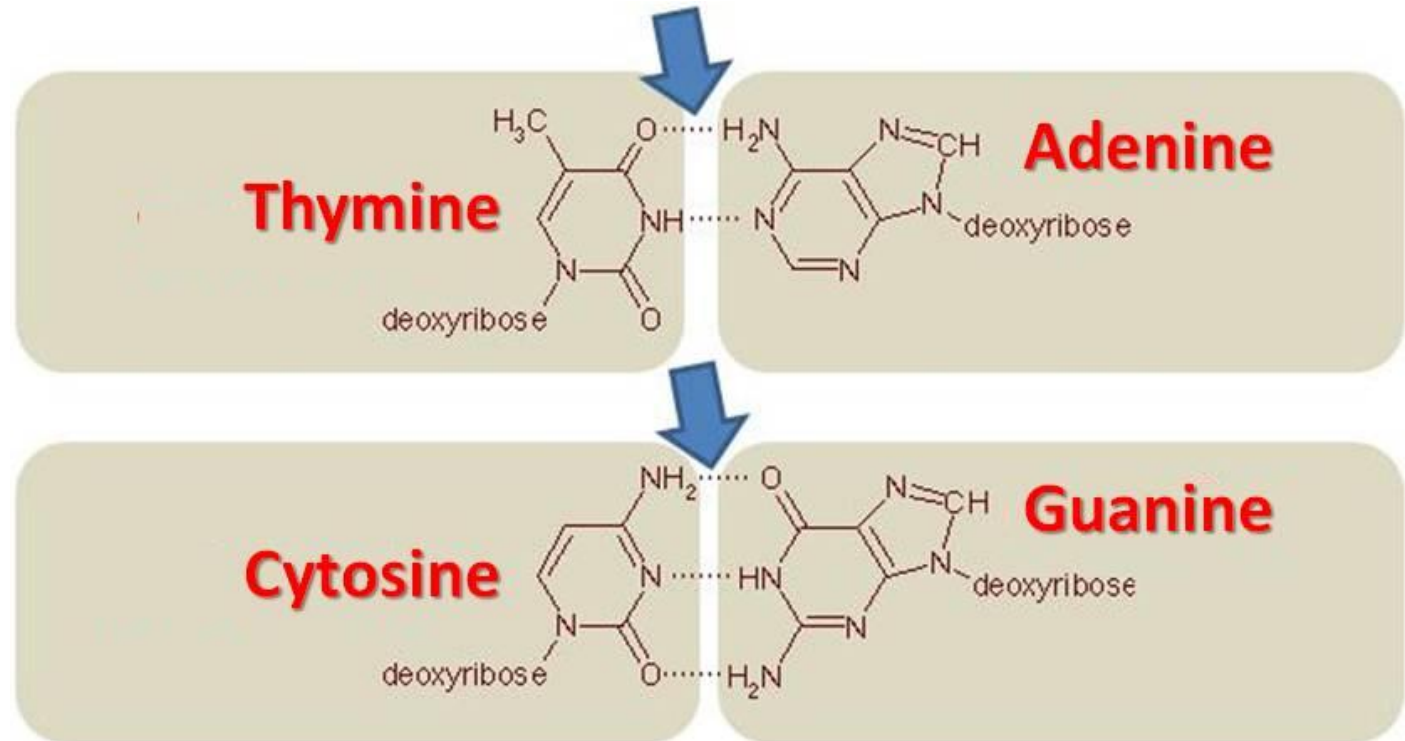
Fig- Structure of purines and pyrimidines

Complementary Base Pairs

In DNA, adenine (A) and thymine (T) are **complementary base pairs**, and cytosine (C) and guanine (G) are also complementary base pairs.

The base pairs are stabilized by **hydrogen bonds**;

- adenine and thymine form two hydrogen bonds between them,
- cytosine and guanine form three hydrogen bonds between them.



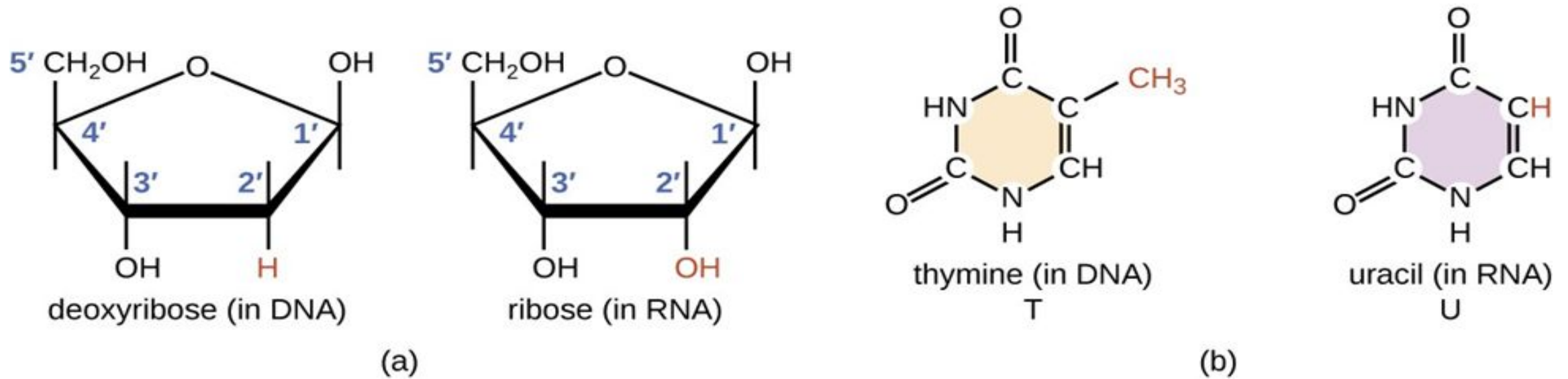


Fig -(a) Ribonucleotides contain the pentose sugar ribose instead of the deoxyribose found in deoxyribonucleotides.

(b) RNA contains the pyrimidine uracil in place of thymine found in DNA.

DNA

1. double-stranded
2. usually found in the nucleus
3. usually more stable
4. typically used for genetic storage

HowBiotech.com

- nucleic acids
- composed of nucleotides
- have a sugar-phosphate backbone
- have nitrogenous bases

RNA

1. single-stranded
2. usually found in the cytoplasm
3. less stable
4. RNA is involved in protein synthesis



THANK YOU